

Final Project: First Deliverable

School of Professional Studies

**Visualization, Feedback and Dissemination - 120 (Spring
2026)**

Group 7

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Due Date: February 1, 2026



1. Selection of a Current Phenomenon or Event

Affordable housing is a critical economic and social issue facing the United States, particularly in metropolitan and micropolitan regions. Housing costs and rental rates have been increasing, while income growth has been uneven in these regions during the past decade. The economic challenges are compounded by the effects of the recent pandemic, economic changes, inflation, and labor market changes, which have been experienced more by low- and middle-income families compared to those in the upper income category.

Another factor contributing to affordable housing is the widening gap between income levels and housing costs, which highlights the need to explore the link between income levels and housing characteristics. This information is critical in order to address the concerns of policymakers and social service organizations seeking to improve affordable housing and the economic situation of individuals and families on a local, state, and national level.

Analyzing these trends will provide insights into income levels and housing characteristics, which will be essential in developing positive changes in various regions of the United States.

2. Dataset Search and Selection

The purpose of this research is to analyze the relationship between income levels and housing affordability. To accomplish the research purpose, two products from the American Community Survey (ACS) published by the United States Census Bureau will be utilized, namely, the "ACS 1-Year Estimates Subject Table S1901: Income in the Last 12 Months" and "ACS 5-Year Estimates Data Profile DP04: Selected Housing Characteristics." The justification for choosing these data sets is based on the fact that they meet the requirements set for conducting the research, namely, the following conditions:

- **Temporal Perspective:** These data sets allow for annual estimates, hence enabling the understanding of the economic situation in the country at any given time.
- **Quantitative and Qualitative Variables:** These data sets have quantitative and qualitative variables. Quantitative variables include income levels, income brackets, housing costs, and the number of rooms in houses, while qualitative variables include tenure type, housing structure type, and occupancy status.
- **Public Accessibility:** These data sets are accessible to the public through the Census Bureau Data Portal in CSV format.

Hence, the S1901 data set provides information regarding income distribution in different regions, while the DP04 data set provides information regarding housing characteristics in different regions.

3I. Goal of the Project (SMART Framework)

The goal of this project is to analyze the relationship between household income distribution and housing characteristics across metropolitan and micropolitan statistical areas in the United States using data from the American Community Survey, in order to gain insight into housing affordability, stability, and regional differences.

In this research project, we will investigate the ways that household incomes are distributed and how housing may differ based on household incomes for Metro and Micro statistical areas in the United States, as reported in the American Community Survey (ACS), and how the correlation between income levels and housing conditions can provide insight about the affordability, stability of housing, and the differences in affordability by region.

The objectives of this project are:

- Specific: Examine income distribution and housing characteristics in metro/micro areas
- Measurable: Income brackets, housing cost burden, tenure type
- Achievable: Uses publicly available ACS datasets
- Relevant: Supports housing policy and planning decisions
- Time-bound: Uses most recent ACS 1-year and 5-year estimates

By integrating income distribution (ACS S1901) with housing characteristics (ACS DP04), this project moves beyond descriptive reporting and enables identification of structural affordability pressures affecting different income groups across metropolitan and micropolitan areas

3II. Research Questions

- How do the income distributions of households differ in metropolitan and micropolitan areas of the United States?
- Which housing characteristics, such as tenure, housing costs, and overcrowding, are most closely associated with income levels?
- Do lower-income households face housing cost burdens in different regions of the country?
- How do housing characteristics differ for renters and owners of homes by income level?
- What regional differences can be identified in comparing income and housing characteristics over time?

3III. Description of the Selected Dataset and Variables

The datasets used in this project are:

- ACS S1901: Income in the Past 12 Months (1-Year Estimates)
- ACS DP04: Selected Housing Characteristics (5-Year Estimates)

These datasets provide both numerical and categorical variables and are publicly accessible through the U.S. Census Bureau.

Variable Name	Type	Range of Values	Why This Variable Is Included
Median Household Income	Numerical	Dollar values (USD)	Core indicator of economic well-being and income distribution
Income Bracket (% of households)	Categorical	Income ranges (e.g., <\$10k, \$50k–\$74k)	Allows comparison of income distribution across regions
Housing Tenure	Categorical	Owner-occupied / Renter-occupied	Helps analyze differences in housing stability
Monthly Housing Costs	Numerical	Dollar values (USD)	Key measure of housing affordability
Housing Cost Burden	Numerical	Percentage (%)	Identifies households spending >30% of income on housing
Geographic Area	Categorical	Metro/Micro areas	Enables regional comparison

References

Doran, G. T. (1981). There's a S.M.A.R.T. way to write management's goals and objectives. *Management Review*, 70(11), 35.

U.S. Census Bureau. (2024). *American Community Survey 1-Year Estimates: Income in the Past 12 Months (S1901)*. <https://data.census.gov>

U.S. Census Bureau. (2023). *American Community Survey 5-Year Estimates: Selected Housing Characteristics (DP04)*. <https://data.census.gov>